Can an AI Explanation Be Misused? Testing Whether SHAP and LIME Help Patients Game a Clinical Diabetes Risk Model

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Abstract

Explainable AI (XAI) methods such as SHAP and LIME are increasingly required by regulation to accompany machine learning predictions in healthcare. A concern is that feature attributions may also serve as instructions for gaming the model—telling a patient which self-reported inputs to change for a better risk score—so that explanations should be withheld. But a patient can also simply try different answers and watch the score move. We compare these two strategies on a diabetes risk model trained on NHANES public health survey data, where height and weight are recorded both as self-reported and as measured by a nurse while the diabetes label comes from a blood test (HbA1c). Lies are restricted to the size and direction of real self-reporting errors observed in NHANES. We train four models—Random Forest and Logistic Regression, each with nurse-measured or self-reported height/weight—and measure the *explanation equivalence* N^*: how many blind attempts give the same mean score reduction as one explanation-guided lie. In a pre-registered confirmatory test on the NHANES 2021–2023 cycle (n = 1,126 test patients), N^* = 1.42 (95% CI 1.21–1.62) for the primary configuration (Random Forest with SHAP) and 0–1.42 across all configurations: one explanation is worth about one or two ordinary attempts at the model. An exploratory analysis of 2017–2018 and a second dataset (BRFSS, 253,680 respondents) agree (N^* = 0, upper 95% bounds ≤ 1.2). Random-Forest lies suggested by an explanation often raise the score instead. The model that uses nurse-measured inputs cannot be gamed and is as accurate. Withholding explanations is therefore not an effective safeguard; verifying self-reported inputs is. We disclose that our first pre-registered test set was examined before the analysis was corrected, and release code, data pipeline and both analysis plans.

CCS Concepts

• **Computing methodologies** → **Machine learning**; • **Social and professional topics** → *Medical records.*

Keywords

explainability, SHAP, LIME, adversarial, gaming, diabetes, NHANES, clinical AI, transparency, fairness

1 Introduction

The deployment of machine learning models in high-stakes clinical settings has prompted widespread calls for transparency. Regulations such as the EU AI Act and GDPR's "right to explanation" push developers to provide explanations alongside predictions [7]. In diabetes risk prediction, a model might inform a patient: "Your risk score is 0.72. Your weight is the most important contributing factor."

This transparency creates a potential dual-use risk: the same explanation that helps a clinician understand the model may help a patient *game* it by adjusting what they self-report. A patient told that weight is the top risk factor might under-report their weight at the next visit. If explanations act as instructions for cheating, one response is to hide them.

Our Question. A patient does not need an explanation to game a model: they can try a few different answers and keep the one with the best score. The relevant question is therefore not whether an explanation helps at all, but whether it helps *more than ordinary trial-and-error*. If explanations add little over a few blind attempts, restricting them is not an effective safeguard. If they add a lot, the benefit of transparency must be weighed against the gaming risk.

Contributions.

(1) We introduce *explanation equivalence* (N^*): the number of blind attempts required to achieve the same mean score reduction as one explanation-guided attempt.
(2) We bound the adversary's lies by *real* self-reporting behaviour: NHANES records both self-reported and nurse-measured height and weight, so we take the size and direction of a realistic lie from the observed gap.
(3) We train two models that differ *only* in whether height/weight are self-reported or measured, isolating the effect of input verification.
(4) We conduct a *pre-registered* confirmatory evaluation of SHAP and LIME on an unseen NHANES cycle and find that, under realistic lies, one explanation is worth about one to two blind attempts ($N^* \leq 1.42$), replicated on a second survey.

2 Related Work

Explainability Methods. Local explanation methods such as LIME [14] and SHAP [11] produce feature attributions that identify the most important inputs for a given prediction. These methods are increasingly embedded in clinical decision support systems [1]. Explanations can look convincing yet be disconnected from the model [2], and Bilodeau et al. [4] proved that for broad model classes feature attributions can do no better than random guessing at predicting how the model responds to a change in its input—showing that a problem is possible, but not how large it is in practice.

Adversarial XAI.. Slack et al. [16] demonstrated that SHAP and LIME can be manipulated at the *model* level to produce misleading explanations, and Bordt et al. [5] argue that post-hoc explanations

suit legal transparency rules poorly because providers can manipulate them. Our work is complementary: we study gaming at the *user* level, asking whether an honest model's explanations help a user game their score. Milli et al. [12] showed that a few explanation queries can replace many ordinary queries for reconstructing a model, and Shokri et al. [15] showed that explanations can leak private training data; both concern the model or its data rather than a patient changing their own answers.

Strategic Classification. Ghalme et al. [6] found that hiding the decision rule usually does not help the decision-maker, and Hikri and Rosenfeld [8] extended this to partial disclosure. König et al. [10] note that advice can improve a prediction without improving the person, just as lying about weight is not losing weight. Unlike these works, we bound manipulation by real human misreporting and compare explanations against an equal-cost blind baseline.

Self-Reporting Bias. NHANES [9] systematically collects both self-reported and nurse-measured anthropometrics, enabling a direct empirical measurement of how people misreport.

3 Problem Setup

3.1 Threat Model

We model an adversary (patient) who:

- Receives a diabetes risk score $f(\mathbf{x}) \in [0, 1]$ from a clinical AI system and wishes to lower it.
- Can change only *self-reported* features $G = \{\text{height}, \text{weight}\}$; nurse-measured values and lab results cannot be changed.
- Lies only *realistically*: in the direction people actually misreport, by at most the lie budget $B(g)$.
- Either receives one SHAP/LIME explanation, or receives none and makes N blind attempts, keeping the best score.

3.2 Realistic Lies

For each gameable feature g, the lie budget is

$$B(g) = \text{Percentile}_{95}\big(|x_g^{\text{self}} - x_g^{\text{measured}}|\big),$$

computed on the 2015–2016 development cycle: $B(\text{height}) = 7.28\,\text{cm}$ and $B(\text{weight}) = 8.96\,\text{kg}$. People over-report height (mean $+1.62\,\text{cm}$) and under-report weight (mean $-0.87\,\text{kg}$), so a realistic lie *increases* height or *decreases* weight. Each lie has a size drawn uniformly from $(0, B(g)]$; the same rule is used by both strategies, so the explanation contributes only *information* (which feature, which direction), never a larger lie.

3.3 Explanation Equivalence

Definition 3.1 (Explanation Equivalence N^*). Let R_{exp} be the score reduction from one explanation-guided lie, averaged over patients, and let $R_{\text{blind}}(N)$ be the reduction from the best of N blind lies, averaged over patients. The *explanation equivalence* is

$$N^* = \inf\{N \geq 0 : R_{\text{blind}}(N) \geq R_{\text{exp}}\},$$

estimated by linear interpolation on $N \mapsto R_{\text{blind}}(N)$ for

$$N \in \{1, 2, 3, 5, 10, 15, 20, 30, 50, 100\}.$$

$N^* = 0$ if one explanation does no better than a single blind attempt; $N^* = 101$ if it beats 100 attempts (censored).

This is the pre-registered primary metric: the average is taken over patients *before* interpolating. As secondary metrics we also report the per-patient N^* distribution.

4 Methodology

4.1 Dataset

We use the NHANES public health survey [9] and merge the demographics (DEMO), body measures (BMX), weight history questionnaire (WHQ), glycohaemoglobin (GHB), blood pressure (BPX), smoking (SMQ), and physical activity (PAQ) files for the 2015–2016 (development), 2017–2018 (exploratory test) and August 2021–August 2023 (confirmatory test) cycles. Refused and don't-know codes are removed per variable as in the NHANES codebooks. Retaining adults (≥18 years) with complete records for both feature sets gives 4,889 participants in 2015–2016 (12.4% with diabetes, HbA1c ≥ 6.5% [3]) and 4,431 in 2017–2018 (13.6%). Within each cycle we use a stratified 80/20 split. Models are *retrained* on each cycle's training split with the locked hyperparameters; only the lie budget (§3.2) carries over from 2015–2016. The 2017–2018 evaluation uses its $n = 887$ test patients; because it was seen before the code was corrected (section 4.6), we treat it as *exploratory* and re-test the final analysis on the August 2021–August 2023 cycle, which no one on the team had analysed (*confirmatory*; section 4.6). In that cycle systolic blood pressure is measured with an oscillometric device (`BPXOSY1`) and vigorous activity is derived from the redesigned activity question (`PAD810Q` > 0); both mappings were fixed before the data was processed.

Supporting datasets. As planned, we use two UCI datasets. UCI Heart Disease (Cleveland, 297 patients) serves as a hand-checkable test of the code: with a logistic regression every risk score, SHAP value, suggested lie and blind best-of-N has a closed form, and the pipeline reproduces all of them to machine precision. The CDC Diabetes Health Indicators dataset (BRFSS 2015, 253,680 respondents) is a comparison: its BMI is computed from self-reported height and weight, so it supports only a Model B analysis, and its diabetes label is itself self-reported.

4.2 Models

Each model uses seven features: age, sex, systolic blood pressure, smoking (ever 100 cigarettes), vigorous recreational activity, height, and weight. *Model A* uses nurse-measured height/weight; *Model B* uses the self-reported values; everything else is identical. We train a Random Forest (RF; 100 trees, maximum depth 10) and a Logistic Regression (LR; standardised inputs) for each, with balanced class weights to account for the 13% prevalence (table 1). On the 2017–2018 test set the models reach AUC 0.77–0.80 and recall 0.69–0.78; Models A and B perform almost identically.

4.3 Explanations

SHAP uses TreeExplainer for RF and LinearExplainer for LR (background: a seeded random sample of 100 training rows). LIME uses its tabular explainer with 500 samples, its default discretisation of continuous features, and all seven features shown.

Table 1: Model Performance on NHANES 2017–2018 Test Set

Config	Input	Model	Accuracy	Precision	Recall	F1	AUC-ROC
measured_rf	measured	RF	0.7396	0.3007	0.686	0.4181	0.7981
measured_lr	measured	LR	0.6652	0.2569	0.7686	0.3851	0.7734
reported_rf	reported	RF	0.7407	0.3032	0.6942	0.4221	0.7933
reported_lr	reported	LR	0.6685	0.2604	0.7769	0.39	0.7796

4.4 Adversarial Strategies

Strategy 1 (Explanation-Guided). The patient requests one explanation and reads it the way it is presented: features ranked by importance, each pushing the risk up or down. Going down the ranking, the patient takes the first gameable feature whose risk-lowering direction is a realistic lie. A SHAP value measures a feature's effect relative to the average patient, so a positive value ("your height raises your risk") means moving the value *towards* the average and a negative one moving it further away; LIME is read from its bins, e.g. raise the feature if the explanation says "*weight* $\leq$ *62 kg* raises risk". If no realistic lie is suggested, the patient does not lie. Otherwise they lie once, with a size drawn as in §3.2, and see the new score; like the blind patient, who keeps the best score so far, they keep their true answers if the lie made the score worse.

Strategy 2 (Blind Trial-and-Error). The patient makes N sequential attempts. Each picks a gameable feature at random and lies in its realistic direction with a size drawn as in §3.2; the best score so far is kept. The simulation runs once up to $N = 100$ and records the running best at each milestone, so blind performance at large N is not over-estimated.

Every patient has its own fixed random seed, so the SHAP and LIME strategies, and Models A and B, are compared against identical blind attempts.

4.5 Statistical Analysis

For each configuration we report N^* with a 95% CI from 1,000 bootstrap resamples of patients; a one-sided paired Wilcoxon signed-rank test [17] of the per-patient explanation reduction against the best of $N = 10$ blind attempts (H_1: explanation > blind), with Holm-adjusted p-values across the eight configurations as a secondary analysis; and Cohen's d. For the Model A vs. B comparison we report the AUC difference on the same test patients with a paired bootstrap CI. All models were implemented with scikit-learn [13].

4.6 Pre-Registration and Deviations

The analysis plan (hypotheses, metric, tests, interpretation thresholds) was recorded as the git tag `v0.1-preregistration` in the project repository on 28 October 2026, before the 2017–2018 data was processed. **A first final run on the held-out data was then made with the code as pre-registered, and its results were seen before the analysis reported here.** That run reported a per-patient mean N^* of 12.4 (95% CI 10.5–14.5) for the primary configuration (reported RF+SHAP) and 9.0–23.1 across all eight configurations, which the pre-registered thresholds would classify as a moderate to high threat. A subsequent code review found implementation errors, which we corrected so that the code matches the plan (primary N^* computed from the averaged curve; the pre-registered 2015–2016 lie budget; only self-reported values gameable, whereas the measured model had been attacked through its nurse-measured inputs). We also made eight changes that *deviate* from the plan: lies only in the realistic direction; the same lie-size rule for both strategies (previously the explanation always used the maximum lie while blind attempts drew smaller ones); LIME showing all seven features (previously five); balanced class weights; a random SHAP background; per-patient random seeds; letting the explanation-guided patient undo a lie that raised their score, as the blind patient implicitly can; and reading the sign of a SHAP value relative to the average patient (the earlier rule told, e.g., shorter-than-average patients to claim to be even shorter; this bug was found by the UCI hand check). Because these settings were chosen after a held-out result had been seen, the 2017–2018 evaluation is not a clean confirmatory test, and the change in conclusion should be read with that in mind. Each deviation is a documented switch in the code; the full timeline and the first run's results are in `docs/preregistration_disclosure.md`.

Confirmatory pre-registration. To obtain a clean test, the corrected analysis, including our predicted result, was pre-registered again (`docs/analysis_plan_v5_confirmatory.md`, git tag `v0.2-preregistration`, pushed to GitHub before the data was processed) and run once, unchanged, on NHANES 2021–2023.

5 Results

5.1 Reporting Gap

Figure 1 shows self-reported minus nurse-measured values. In 2017–2018, participants over-report height by +1.69 cm on average (SD 3.23) and under-report weight by −0.67 kg (SD 4.84), closely matching 2015–2016 (+1.62 cm, SD 3.41; −0.87 kg, SD 4.79). The 95th-percentile gap in 2017–2018 (6.88 cm, 9.32 kg) is within 0.4 cm and 0.4 kg of the development budget we use.

5.2 Main Result: Explanation Equivalence (2017–2018, exploratory)

On the 2017–2018 test set (exploratory, see section 4.6), $N^* = 0.0$ in all eight configurations (table 2, fig. 2); the primary configuration (Model B, RF+SHAP) has a 95% CI of 0.0–1.2. For Model B, one explanation-guided lie lowers the risk score by 0.018–0.024 on average, a single blind attempt by 0.024–0.025, and the best of 10 blind attempts by 0.046–0.064 (table 3): an explanation is worth about one ordinary attempt at the model, never more. The confirmatory result on 2021–2023 is in section 5.9.

Table 2: Main Results: Explanation Equivalence N^* Across All 8 Configurations

Input	Model	Expl.	N patients	N^*	95% CI	Per-patient mean N^*	Per-patient median N^*	% Useless	% Worth 10+
Reported	RF	SHAP	887	0.0	[0.00, 1.22]	7.8	0.0	69.6%	9.5%
Reported	RF	LIME	887	0.0	[0.00, 0.00]	7.07	0.0	73.8%	8.1%
Reported	LR	SHAP	887	0.0	[0.00, 1.09]	3.61	0.0	52.3%	6.4%
Reported	LR	LIME	887	0.0	[0.00, 0.00]	2.99	0.0	56.3%	6.1%
Measured	RF	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%

Table 3: Mean risk-score reduction (0-1 scale): one explanation-guided lie vs. the best of N blind lies, and how often the explanation suggested no lie or a lie that raised the risk (which the patient then undoes; the column "lie not undone" shows the v3 rule)

Input	Model	Expl.	1 explanation	1 expl. (lie not undone)	1 blind	10 blind	100 blind	% no lie suggested	% lie raised risk (undone)
Reported	RF	SHAP	0.0233	0.0177	0.024	0.0638	0.0785	1.8%	28.5%
Reported	RF	LIME	0.0181	0.0117	0.024	0.0638	0.0785	3.3%	31.1%
Reported	LR	SHAP	0.0242	0.0242	0.0248	0.0457	0.0525	0.0%	0.0%
Reported	LR	LIME	0.0223	0.0223	0.0248	0.0457	0.0525	7.7%	0.0%
Measured	RF	SHAP	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%

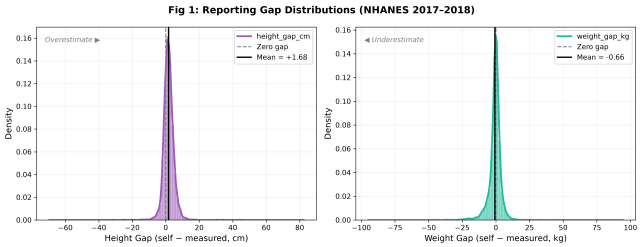


Figure 1: Self-reported minus nurse-measured height and weight (NHANES 2017–2018). Height is over-reported and weight under-reported.

5.3 Why the Explanation Does Not Help More

With realistic lies the direction is fixed (taller, lighter) and the lie size is drawn the same way for both strategies, so an explanation can only tell the patient *which* of height and weight to change; two or three blind attempts reveal the same. For the Random Forest the suggested lie also often backfires: 28.5% (SHAP) and 31.1% (LIME) of explanation-guided lies *raise* the score and are undone, because local attributions do not describe the model's response over the whole range of a lie, the practical face of the impossibility result of Bilodeau et al. [4]. For the Logistic Regression, whose response is monotone, no SHAP-guided lie backfires. The two late corrections (undoing a bad lie; reading SHAP relative to the average patient) raised the primary configuration's explanation reduction from 0.013 to 0.018 to 0.023, against 0.024 for one blind attempt, without changing N^* (`docs/preregistration_disclosure.md`).

5.4 Per-Patient Distribution

Per patient (fig. 3), the explanation is no better than one blind attempt for 69.6% of patients (RF+SHAP) and worth 10 or more attempts for 9.5% (6.7% beat all 100 attempts). The per-patient mean N^* (7.8 for RF+SHAP, 3.6 for LR+SHAP) is therefore driven by a small tail.

5.5 Model A vs. Model B

That Model A cannot be gamed is true *by design*: its height and weight come from the nurse, so it has no input the patient can change, and its zeros in table 4 follow from the threat model rather than from the experiment. The measurable comparison has two parts. Model B, which differs only in using self-reported values, can be lowered by 0.046–0.064 with ten blind attempts and 0.053–0.079 with a hundred (fig. 4). And using measured instead of self-reported inputs changes accuracy by an amount indistinguishable from zero on the same test patients (table 5): AUC difference +0.005 for RF (95% CI −0.009 to +0.018) and −0.006 for LR (−0.013 to +0.000). Verifying the two self-reported inputs removes the attack surface at no measurable cost in accuracy.

5.6 SHAP vs. LIME Agreement

Because each patient receives the same lie size under SHAP and LIME, differences come from the explanation alone (fig. 5). The two methods lead to the same outcome for 74% (RF) and 71% (LR) of patients, with per-patient correlation $r = 0.65$ and 0.87. Both give $N^* = 0$; SHAP is slightly better than LIME (RF 0.023 vs. 0.018; LR 0.024 vs. 0.022).

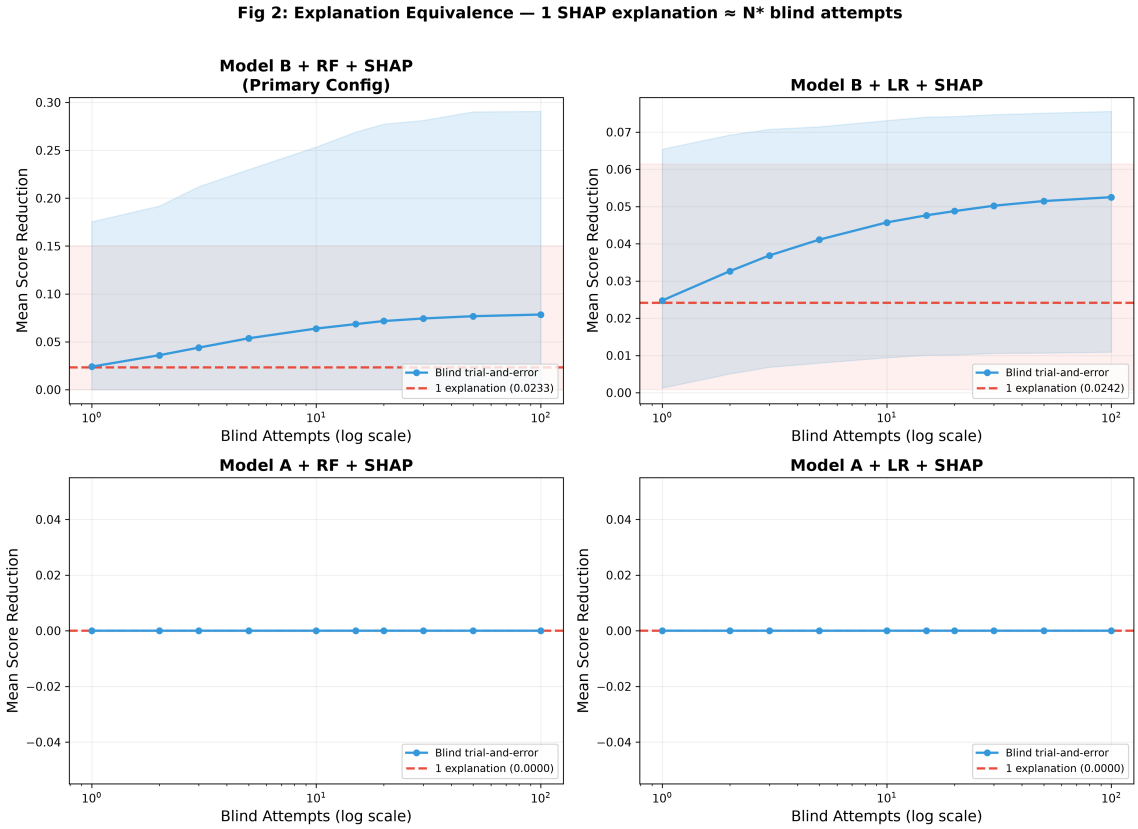


Figure 2: Mean score reduction from the best of N blind attempts (blue) vs. one explanation-guided lie (red dashed), SHAP, 2017–2018. For Model B the explanation line does not rise above the first blind attempt, so $N^* = 0$. Model A cannot be gamed.

Table 4: Model A (Measured) vs Model B (Self-Reported): Vulnerability. Model A has no input the patient can change, so its zeros are true by design, not an experimental finding (see Table 6b).

Config	Model A N^*	Model B N^*	A: reduction, 10 blind	B: reduction, 1 explanation	B: reduction, 10 blind	B can be gamed
RF+SHAP	0.0	0.0	0.0	0.0233	0.0638	Yes
RF+LIME	0.0	0.0	0.0	0.0181	0.0638	Yes
LR+SHAP	0.0	0.0	0.0	0.0242	0.0457	Yes
LR+LIME	0.0	0.0	0.0	0.0223	0.0457	Yes

Table 5: Self-reported vs measured height/weight on the same patients: accuracy difference (Model A minus Model B, paired bootstrap 95% CI) and how far Model B can be gamed. Model A cannot be gamed by construction.

Model	AUC A (measured)	AUC B (self-reported)	Difference A-B	95% CI (paired bootstrap)	B: max reduction, 10 blind lies
RF	0.7981	0.7933	0.0047	[-0.0093, +0.0177]	0.0638
LR	0.7734	0.7796	-0.0062	[-0.0127, +0.0002]	0.0457

5.7 Statistical Tests

The one-sided Wilcoxon test of explanation > blind at $N = 10$ is not significant in any configuration ($p = 1.0$, also after Holm correction; table 6). The effect sizes are large and *negative* ($d = -0.66$ to -1.24 for Model B): ten blind attempts lower the score clearly more than one explanation.

5.8 Comparison Dataset: CDC Diabetes Health Indicators

On 1,000 sampled test respondents of BRFSS 2015 (UCI 891), with self-reported BMI as the only gameable feature, the result is the same (table 7): $N^* = 0.0$ in all four configurations (upper CI bound

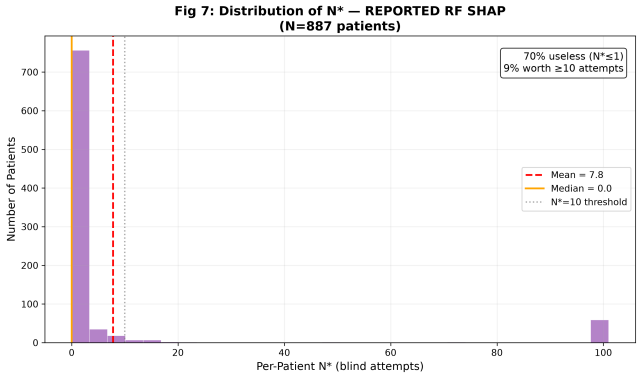


Figure 3: Distribution of per-patient N^* (Model B, RF + SHAP; $n = 887$ patients, 2017–2018).

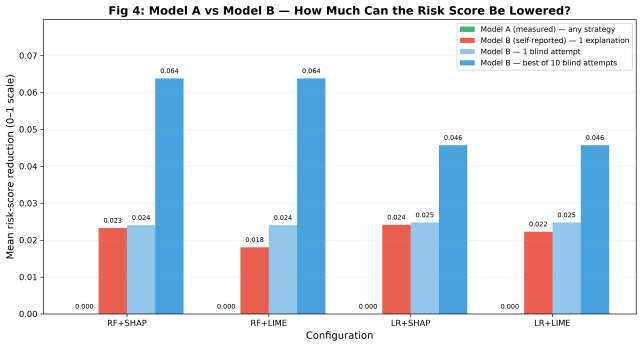


Figure 4: Mean risk-score reduction for Model A (nurse-measured inputs) and Model B (self-reported inputs).

Table 6: Statistical Tests: Explanation vs Blind (N=10), Wilcoxon Signed-Rank

Config	Wilcoxon p	p (Holm)	Cohen's d	Significant
reported_rf_shap	1.000000	1.000000	-0.659	No
reported_rf_lime	1.000000	1.000000	-0.720	No
reported_lr_shap	1.000000	1.000000	-1.196	No
reported_lr_lime	1.000000	1.000000	-1.238	No
measured_rf_shap	1.000000	1.000000	0.000	No
measured_rf_lime	1.000000	1.000000	0.000	No
measured_lr_shap	1.000000	1.000000	0.000	No
measured_lr_lime	1.000000	1.000000	0.000	No

at most 1.1); one explanation-guided lie lowers the score by 0.026–0.041 against 0.037–0.044 for one blind attempt. BRFSS has no measured BMI and a self-reported diabetes label, so this supports only the Model B finding, with a BMI lie budget borrowed from NHANES.

5.9 Confirmatory Result: NHANES 2021–2023

The confirmatory run used the code and settings fixed at the `v0.2-preregistration` tag (commit `5b2f240`), unchanged, on NHANES August 2021–August 2023: 5,628 adults with complete data (11.9% with diabetes) and 1,126 test patients. The reporting gap was again stable (height +1.73 cm, weight −0.65 kg on average; 95th percentiles 6.84 cm and 9.04 kg against the 7.28 cm and 8.96 kg budget used).

For the primary configuration, $N^* = 1.42$ (95% CI 1.21–1.62): one SHAP explanation is worth about one and a half blind attempts, inside the pre-registered "minimal advantage" band ($N^* < 5$), and the upper confidence bound is far below 5 (table 8). Across Model B, N^* ranges from 0 to 1.42; the one-sided Wilcoxon test of explanation $>$ ten blind attempts is not significant in any configuration ($p = 1.0$, also after Holm correction), with large negative effect sizes. Every pre-registered prediction held. As in 2017–2018, Random Forest lies suggested by an explanation backfired for 27% of patients. Unlike 2017–2018, SHAP now does slightly *better* than one blind attempt (0.034 vs. 0.028 for the Random Forest), which is why N^* is above zero; LIME does not.

Model A again cannot be gamed. The accuracy difference between measured and self-reported inputs on the same test patients is +0.006 for the Random Forest (95% CI −0.014 to +0.026) and −0.012 for the Logistic Regression (−0.023 to −0.002), i.e. in this cycle the self-reported Logistic Regression is slightly *more* accurate. Either way the difference is about one hundredth of AUC, small against the 0.06–0.07 by which ten blind lies lower a Model B score.

6 Discussion

6.1 Policy Implications

Our pre-registered interpretation guide set three bands: $N^* < 5$ (withholding explanations is not a meaningful safeguard), $5 \le N^* \le 20$ (moderate advantage), and $N^* > 20$ (explanations are powerful attack vectors). With $N^* \le 1.42$ in every configuration of the confirmatory test (upper CI bound 1.62), the result falls in the first band. Under realistic lies, an explanation does not give a patient a better route to a lower score than simply trying an answer; withholding it would remove information from honest users while barely slowing a motivated one. The vulnerability lies in the self-reported inputs: the measured-input model cannot be gamed, and the two models are equally accurate.

This conclusion runs against the intuition that explanations are "instructions for cheating". The reason is that a feature attribution answers "what contributed to this score?", not "what should I change?", and the two can point in different directions.

6.2 Limitations

- **Model of the patient.** We model one way of reading an explanation (follow the top realistic suggestion once). A more sophisticated reader, or a counterfactual explanation, could extract more.
- **Single lie, single feature.** Each explanation-guided attempt changes one feature once; multi-feature lies could be more powerful for both strategies.
- **Lie model.** Lie size is uniform within the 95th-percentile gap and in the observed direction, so an explanation can only choose *which* feature to change. Using the maximum lie, or allowing either direction, is a one-line change in our code; we leave that sensitivity analysis to future work.

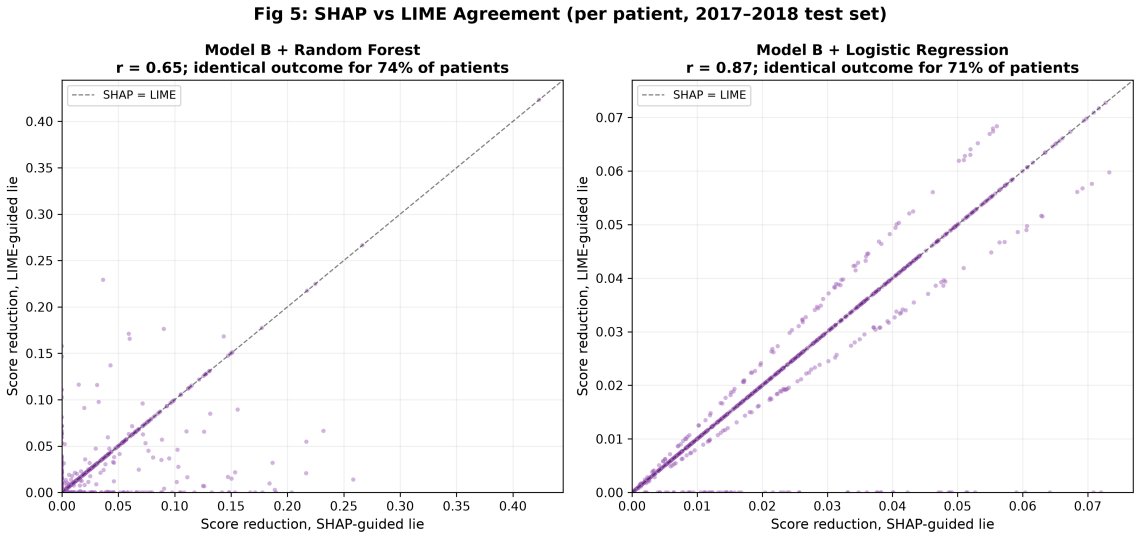


Figure 5: Per-patient score reduction from the SHAP-guided lie vs. the LIME-guided lie (Model B, 2017–2018 test set).

Table 7: Comparison dataset (CDC Diabetes Health Indicators, UCI 891; BRFSS 2015, 1,000 sampled test respondents): one explanation vs. blind attempts, with self-reported BMI as the only gameable feature.

Model	Expl.	N	N*	N* 95% CI	1 explanation	1 blind	10 blind	Per-patient mean N*	% useless	% lie undone	Wilcoxon p	Cohen's d
RF	SHAP	1000	0.0	[0.00, 1.04]	0.0409	0.0436	0.0772	16.98	67.8	16.5	1.0	-0.703
RF	LIME	1000	0.0	[0.00, 0.00]	0.0324	0.0436	0.0772	13.86	74.7	13.6	1.0	-0.829
LR	SHAP	1000	0.0	[0.00, 1.12]	0.0369	0.0373	0.0655	4.04	49.8	0.0	1.0	-1.142
LR	LIME	1000	0.0	[0.00, 0.00]	0.0257	0.0373	0.0655	2.85	63.8	0.0	1.0	-1.316

Table 8: Confirmatory result, NHANES 2021–2023 (pre-registered, $n = 1{,}126$ test patients). Mean score reductions; Model A (measured) is 0 in every column by design.

Model B config	N^*	95% CI	1 expl.	1 blind	10 blind	Cohen's d
RF + SHAP (primary)	1.42	[1.21, 1.62]	0.034	0.028	0.070	−0.71
RF + LIME	0.00	[0.00, 0.00]	0.024	0.028	0.070	−0.76
LR + SHAP	1.07	[0.00, 1.20]	0.035	0.035	0.062	−1.14
LR + LIME	0.00	[0.00, 0.00]	0.032	0.035	0.062	−1.19

- **Domain.** Results are for diabetes risk from anthropometric self-report (NHANES, and BRFSS as a comparison whose diabetes label is itself self-reported).
- **Survey changes.** In 2021–2023 blood pressure is measured oscillometrically and the activity question was redesigned; we mapped both before seeing the data.
- **Pre-registration and deviations.** The plan was recorded as a git tag in the project repository rather than an independent registry; a first run on the 2017–2018 data was seen before the analysis was corrected, and eight settings deviate from the first plan (section 4.6). We therefore treat 2017–2018 as exploratory and rest our conclusion on the re-registered 2021–2023 test.

6.3 Recommendations for Practitioners

(1) Prefer **verified inputs** (e.g. nurse measurement) over self-report for the features that drive high-stakes predictions.
(2) Do not withhold explanations as an anti-gaming measure; audit the gain they give with the N^* framework.
(3) Be aware that attributions are not advice: present them as "contributions" rather than "what to change".

7 Conclusion

We introduced the *explanation equivalence* metric N^* and applied it to a diabetes risk model using real NHANES survey data, with lies bounded by how people actually misreport. In a pre-registered confirmatory test on unseen NHANES 2021–2023 data, one SHAP or LIME explanation was worth at most about one and a half blind

attempts ($N^* = 1.42$, 95% CI 1.21–1.62, for the primary configuration; 0–1.42 overall), and an exploratory 2017–2018 analysis and a second survey agreed. For the Random Forest, explanation-guided lies often raised the score instead. A model using nurse-measured inputs could not be gamed and was as accurate. The effective safeguard against gaming is therefore verifying self-reported inputs, not withholding explanations.

Future Work. Sensitivity to the lie model, multi-feature and multi-step attacks, counterfactual explanations, other clinical domains, and a user study of how real patients read explanations.

Code and Data. Code, analysis plan and results: https://github.com/AkkiNikumbh/BTECH_PROJECT. NHANES data are public and are downloaded by the pipeline.

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